

INDIAN STATISTICAL INSTITUTE, KOLKATA

End Semester Exam , Second semester 2024-25

Algebra 2 , M.Math 1

Full Marks : 50

Time : 3 hours

Date : 02/05/2025

1. Let G be a finite group such that $|G|$ is odd. Let $1 < d < |G|$ be a divisor of $|G|$ such that there exist exactly three subgroups H_1, H_2, H_3 of G of order d . Let $N_G(H_i)$ denote the normalizer of the subgroup H_i and let H_1 be not normal.
 - (a) Show that $\sum_{g \in G} |\{H_i | gH_i g^{-1} = H_i\}| = |G| = \sum_{i=1}^3 |N_G(H_i)|$. [5]
 - (b) Show that $G \neq \cup_{i=1}^3 N_G(H_i)$. [3]
2. Show that any non trivial subgroup $H \subset G$, where $|G| = 2024$, has a non-trivial normal cyclic subgroup. [6]
3. Let p be a prime. Show that a regular p -gon of unit side length can be constructed using ruler and compass if and only if $p = 2^{2^n} + 1$ for some non negative integer n . [5]
4. Show that there exists $K/\mathbb{Q}$ Galois extension and $\mathbb{Q} \subset K' \subset K$ subfield such that $G(K/K')$ is a simple non abelian group. [5]
5. Let $K := \mathbb{Q}(x_1, \dots, x_n)$ be the fraction field of the polynomial ring $\mathbb{Q}[x_1, \dots, x_n]$ and let s_i be the i -th symmetric polynomial in n -variables.
 - (a) Show that $S_n \cong G(K/\mathbb{Q}(s_1, \dots, s_n))$. [5]
 - (b) Show that the polynomial $f(t) := \prod_{i=1}^n (t - x_i)$ is an irreducible polynomial in $\mathbb{Q}(s_1, \dots, s_n)[t]$. [5]
 - (c) Find the fixed field of the subgroup $S_{n-1} \subset S_n$, where S_{n-1} acts on n -letters by permuting the first $n - 1$ letters and leaving the n -th letter fixed. [3]
6. let $1 < n$ be a natural number and let ζ_n be a primitive n -th root of unity.
 - (a) Show that there exists irreducible polynomial $f(x) = x^n - a \in \mathbb{Q}[x]$ such that $f(x)$ does not have any root in $\mathbb{Q}(\zeta_n)$. [5]
 - (b) Show that $\mathbb{Q}(\zeta_n, a^{1/n})/\mathbb{Q}(\zeta_n)$ is a cyclic Galois extension. [3]
 - (c) Deduce that there exist $b \in \mathbb{Q}(\zeta_n)$ and some $m > 1$ natural number such that $x^m - b$ is irreducible in $\mathbb{Q}(\zeta_n)[x]$ and $\mathbb{Q}(\zeta_n, a^{1/n})$ is splitting field of $x^m - b$ over $\mathbb{Q}(\zeta_n)$. [5]

$$2024 = 8 \times 253 + 8 \times 11 \times 23$$